

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

CN Lecture 06: DHCP (Dynamic Host Configuration Protocol) and ARP (Address Resolution Protocol)

OBE Based Questions and Answers

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1. A newly installed computer is connected to a LAN but has no IP address. Explain how DHCP automatically assigns an IP address to the computer.

Answer:

DHCP automatically assigns an IP address through the following four-step process:

- (a) The client broadcasts a **DHCPDISCOVER** message.
- (b) The DHCP server responds with a **DHCPOFFER** message containing an available IP address.
- (c) The client selects one offer and sends a **DHCPREQUEST** message.
- (d) The server confirms the allocation by sending a **DHCPACK** message.

After receiving the DHCPACK, the client enters the **BOUND** state and can use the assigned IP address.

2. A network administrator manages a large organization where employees frequently connect and disconnect their laptops. Explain why DHCP is preferred over manual IP configuration.

Answer:

DHCP is preferred because it:

- Automatically assigns IP addresses.
- Reduces manual configuration errors.
- Prevents duplicate IP addresses.
- Supports temporary IP address leasing.
- Simplifies network administration.
- Efficiently manages available IP addresses.

Therefore, DHCP is ideal for networks with frequently changing devices.

3. A company previously used BOOTP but now plans to upgrade to DHCP. Compare BOOTP and DHCP and explain why DHCP is more suitable.

Answer:

Feature	BOOTP	DHCP
Address Assignment	Static	Static and Dynamic
IP Lease	Not Supported	Supported
Address Pool	No	Yes
Temporary IP Assignment	No	Yes
Network Flexibility	Low	High

DHCP is more suitable because it dynamically allocates IP addresses and supports lease renewal.

4. A DHCP client has successfully obtained an IP address. Explain the purpose of the BOUND, RENEWING, and REBINDING states.

Answer:

- **BOUND:** The client uses the assigned IP address during the lease period.
- **RENEWING:** At 50% of the lease time, the client requests lease renewal from the original DHCP server.

- **REBINDING:** At 87.5% of the lease time, if renewal fails, the client requests renewal from any available DHCP server.

These states ensure continuous network connectivity without manual intervention.

5. **A client receives multiple DHCPOFFER messages from different DHCP servers. Explain how the client selects an IP address and completes the configuration process.**

Answer:

The client performs the following steps:

- (a) Receives multiple DHCPOFFER messages.
- (b) Selects one offered IP address.
- (c) Sends a DHCPREQUEST message to the selected server.
- (d) Waits for a DHCPACK message.
- (e) After receiving DHCPACK, the client enters the BOUND state.

The selected server reserves the offered IP address for the client.

6. **A user's computer knows the destination IP address but not the destination MAC address. Explain how ARP enables successful communication.**

Answer:

ARP resolves the destination MAC address as follows:

- (a) The sender broadcasts an ARP Request containing the destination IP address.
- (b) All devices receive the request.
- (c) Only the device whose IP address matches the requested IP responds.
- (d) The destination sends an ARP Reply containing its MAC address.
- (e) The sender stores the mapping in its ARP cache and transmits the data frame.

Thus, ARP maps an IP address to its corresponding MAC address.

7. **Three computers are connected to the same LAN. PC-A wants to communicate with PC-C but knows only PC-C's IP address. Describe the complete ARP communication process.**

Answer:

- (a) PC-A broadcasts an ARP Request using the broadcast MAC address (FF:FF:FF:FF:FF:FF).
- (b) PC-B receives the request but ignores it because the destination IP does not match its own.
- (c) PC-C receives the request and finds that the destination IP matches its own IP address.
- (d) PC-C sends an ARP Reply containing its MAC address directly to PC-A.
- (e) PC-A stores the MAC address in its ARP cache and begins communication.

8. **Explain the importance of the broadcast MAC address in the ARP Request process.**

Answer:

The broadcast MAC address (FF:FF:FF:FF:FF:FF) ensures that every device on the local network receives the ARP Request.

Each device checks whether the requested IP address matches its own.

Only the correct destination replies with its MAC address, while all other devices discard the request.

9. **A network administrator notices that repeated ARP Requests are increasing network traffic. Explain how the ARP cache improves communication efficiency.**

Answer:

The ARP cache temporarily stores IP-to-MAC address mappings.

Advantages include:

- Reduces repeated ARP broadcasts.
- Speeds up communication.
- Decreases network overhead.
- Improves overall network performance.

When an entry expires, it is removed automatically according to the ARP cache timeout.

10. **A client is unable to renew its DHCP lease after reaching 50% of the lease period. Explain what happens next according to the DHCP state transition process.**

Answer:

If the client does not receive a DHCPACK during the RENEWING state:

- The client continues using the assigned IP address.
- When 87.5% of the lease time expires, it enters the REBINDING state.
- The client broadcasts a DHCPREQUEST to any available DHCP server.
- If a DHCPACK is received, the lease is renewed.
- Otherwise, if the lease expires or a DHCPNACK is received, the client returns to the INIT state and requests a new IP address.

11. **Evaluate the advantages and limitations of DHCP in modern computer networks.**

Answer:

Advantages

- Automatic IP address assignment.
- Efficient address management.
- Reduces administrative workload.
- Supports dynamic addressing.
- Prevents IP conflicts.

Limitations

- Depends on DHCP server availability.
- DHCP server failure may prevent new devices from obtaining IP addresses.
- Incorrect server configuration may affect multiple clients.

12. **Evaluate the role of ARP in local area network communication and discuss its advantages and disadvantages.**

Answer:

ARP enables communication between devices by translating IP addresses into MAC addresses.

Advantages

- Automatically resolves MAC addresses.
- Enables successful communication within a LAN.
- Maintains ARP cache for faster communication.

Disadvantages

- Broadcast requests increase network traffic.
- ARP cache entries may become outdated.
- Vulnerable to ARP spoofing attacks.

Therefore, ARP is an essential protocol for LAN communication but requires proper security mechanisms.